

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 16

Category Theory: The Mathematics of Mathematics

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A functor from topology to algebra (like the fundamental group) translates topological questions into algebraic questions. What would it mean for this functor to be “faithful”, i.e., not to lose information? Does the fundamental group lose information?

2. Someone says: “Category theory is just abstraction for its own sake: it doesn’t tell you anything new.” Construct a counterargument using a specific example from this curriculum.

3. The product of two vector spaces $V \times W$ and the product of two groups $G \times H$ are both constructed by the “same” categorical process. What is that process? (State it in terms of universal properties.)

4. You have been studying structures (groups, spaces, fields) and maps between them (homomorphisms, continuous functions, linear maps). Category theory adds a new layer: maps between maps (functors, natural transformations). Why would you need maps between maps?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
